FedKAN-IDS: Heterogeneity-Robust Federated Kolmogorov–Arnold Networks for NetFlow-Based IoT Intrusion Detection

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***Abstract*—Federated Learning (FL) for IoT intrusion detection systems (IDS) faces a fundamental tension between the bandwidth budget of edge devices and the catastrophic accuracy degradation that dense neural network architectures exhibit when client data is heterogeneously distributed. We present FedKAN-IDS, the first systematic study of Federated Kolmogorov–Arnold Networks (KANs) on NetFlow-based IoT intrusion detection, with a contribution that is empirical rather than methodological: we show that under extreme client heterogeneity (Dirichlet $\alpha = 0.1$), the worst-seed binary detection accuracy of FedKAN-8 (~ 3.3k parameters) is 80.7%, while a parameter-matched FedAvg-MLP at ~ 3.4k parameters drops to 56.5% on the same partition. Doubling MLP capacity does not close this gap, indicating an architectural rather than capacity-driven advantage. The seed-paired bootstrap 95% confidence interval for the FedKAN–MLP-PM accuracy difference is $[+0.50, +14.57]$ percentage points across $n = 10$ seeds, and FedKAN's per-class F1 on the DoS attack class is $+8$ percentage points higher than the parameter-matched MLP under multi-class non-IID. We anchor the empirical finding in a convergence analysis showing that the KAN B-spline approximation introduces an additive $\mathcal{O}(G^{-(k+1)})$ bias term that is independent of the data-heterogeneity factor Γ_α, behaving as an effective regulariser. Code, configurations, and 146 reproducible runs are publicly available.**

***Index Terms*—Federated Learning, Kolmogorov-Arnold Networks, IoT Security, Intrusion Detection, Non-IID, Robustness**

I. Introduction

THE pervasive deployment of Internet-of-Things (IoT) devices across smart cities, industrial control, healthcare, and autonomous transport has dramatically expanded the network attack surface. Distributed denial-of-service (DDoS) campaigns, botnets, reconnaissance scans, and other intrusions traverse the same edge gateways that carry legitimate IoT traffic, motivating the deployment of network intrusion detection systems (IDS) at the network edge [1].

Federated learning for IoT IDS. Centralised IDS pipelines, in which raw flow records are exfiltrated to a central server for model training, raise persistent privacy concerns and conflict with regulations such as GDPR and HIPAA. Federated Learning (FL) [2] offers a privacy-preserving alternative: each IoT gateway trains a local model on its own traffic and only the model updates are aggregated. A growing literature applies FL to IoT IDS, with notable contributions including DeepFed [3], which combined CNNs with gated recurrent units, and a wave of subsequent work on heterogeneity-robust aggregation, communication compression [4]–[6], and security-aware client selection.

The communication–robustness tension. Two pressures shape FL-IDS design at the edge. First, IoT links are bandwidth-constrained: a typical LoRaWAN device has a duty-cycled uplink budget on the order of kilobytes per hour. Second, client data is highly non-identical. The geographically distributed nature of IoT deployments means that one gateway may see almost only DDoS traffic while another sees primarily reconnaissance scans, a scenario commonly modelled by a Dirichlet partition with a small concentration parameter α [7]. Under such partitioning, vanilla FedAvg-MLP architectures are known to suffer slow convergence and, more pathologically, occasional catastrophic divergence in which a single unfortunate client–initialisation pair drags the global model far from the consensus optimum.

Kolmogorov–Arnold Networks (KANs). The recent introduction of KANs [8] replaces fixed neuron-wise activations with learnable edge-wise B-spline activations. Unlike a dense MLP weight matrix, where each scalar parameter modulates a global linear interaction, each spline coefficient in a KAN layer has *local* support on the grid by construction of the B-spline basis. This locality has been argued to provide better interpretability and parameter efficiency in physics-discovery and scientific-computing settings.

KAN-FL: a recently busy research line. Within a year of [8], several works have integrated KANs into federated learning. F-KANs [9] ported the original KAN to a vanilla FedAvg framework on CIFAR-style benchmarks. [10] performed the first systematic non-IID evaluation of FedKAN, reporting modest gains over MLPs at matched parameter budgets on tabular data. [11] extended the comparison across multiple aggregation strategies (FedProx, SCAFFOLD), and [12] applied a Fed-KAN to traffic prediction. [13] contributed a medical-imaging benchmark and [14] an ECG-signal study. Most recently, [15] proposed CG-FKAN, which sparsifies the KAN grid coefficients to attack communication efficiency directly.

Positioning of this paper. The methodological observation "KAN-FL converges in fewer rounds and transmits fewer bytes than a parameter-heavy MLP" has therefore been published several times outside the IDS setting. We do *not* claim that observation as new. What is new in this paper is (i) a systematic empirical study of FedKAN *specifically* on the standardised NetFlow-v2 family of IDS datasets [16], (ii) a parameter-matched comparison protocol that resolves the unfair-baseline ambiguity present in much prior FedKAN work, and (iii) a refined narrative anchored in a convergence

analysis: FedKAN's advantage is not in mean accuracy but in worst-case robustness under extreme client heterogeneity, and this advantage cannot be replicated by simply scaling up an MLP baseline.

Headline empirical finding. On NF-BoT-IoT-v2 binary detection under Dirichlet $\alpha = 0.1$ (extreme non-IID) with $K = 10$ clients and $T = 50$ rounds, averaged over $n = 10$ seeds: the parameter-matched FedAvg-MLP-PM (3,362 parameters) collapses to as low as 56.5% accuracy on its worst seed, while FedKAN-8 (3,280 parameters) maintains $\geq 80.7\%$ on every seed, with a $2.6\times$ smaller cross-seed standard deviation. The seed-paired bootstrap 95% confidence interval for the mean accuracy difference is $[+0.50, +14.57]$ percentage points. Doubling MLP capacity to 3.4k parameters *worsens* the worst-seed accuracy from 60.4% (small MLP at 1.3k params) to 56.5%, ruling out a capacity-driven explanation.

Theoretical anchoring. We provide a convergence analysis (Theorem 1) showing that the FedKAN error decomposes into a standard FedAvg term proportional to the data-heterogeneity factor Γ_α *plus* an additive spline-approximation bias of order $\mathcal{O}(G^{-(k+1)})$. Crucially, the spline-bias term is independent of Γ_α, behaving as an effective regulariser whose relative impact *decreases* as heterogeneity grows. Empirically, this manifests as the smaller cross-seed variance we report.

Contributions.

- We present the first systematic, parameter-matched empirical comparison of FedKAN against FedAvg-MLP on the standardised NetFlow-v2 IDS datasets [16], under both binary and 5-class detection and three data-heterogeneity regimes (IID, Dirichlet $\alpha \in \{1.0, 0.1\}$). Our protocol explicitly addresses the parameter-matching gap identified in prior FedKAN work.
- We identify a robustness/tail-risk advantage of FedKAN that is architectural rather than capacity-driven: parameter-matched MLP suffers worst-seed accuracy drops of ~ 24 percentage points relative to FedKAN-8 under extreme non-IID, and doubling MLP capacity does not close this gap. The advantage is supported by a seed-paired bootstrap 95% CI excluding zero across $n = 10$ seeds.
- We anchor the empirical finding in a convergence analysis introducing a spline-bias term $\mathcal{O}(G^{-(k+1)})$ that is independent of the heterogeneity factor and acts as an effective regulariser (Theorem 1 and Corollary 2).
- We honestly walk back the "$\sim 50\%$ communication reduction" framing of prior FedKAN-IDS work, showing via Proposition 2 that under parameter parity FedKAN incurs roughly 14% *more* bytes per round than its MLP counterpart, and reframe the case for FedKAN around tail-risk reduction, the more deployment-relevant metric for IDS.
- All code, dataset preparation scripts, configurations, and 146 reproducible runs are released at https://github.com/haodpsut/FedKAN-IDS-v2.

Organisation. Section II introduces the KAN layer formalism and the federated IDS optimisation problem. Section III details the FedKAN-IDS framework, including server and client algorithms. Section IV presents the four formal results that anchor our empirical narrative. Section V reports experimental protocol and results across architectures, partitions, and modes. Section VI discusses threats to validity and outlines next steps.

II. Preliminaries and Problem Formulation

A. Kolmogorov–Arnold Network Layer

Kolmogorov–Arnold Networks (KANs) [8] replace the fixed neuron-wise activations of an MLP with *learnable* edge-wise activations parameterised as B-splines. Formally:

Definition 1 (KAN Layer)**.** *Given input $\mathbf{x} \in \mathbb{R}^{n_{in}}$, grid size $G \in \mathbb{N}$, and B-spline order $k \in \mathbb{N}$, a KAN layer is the map $\Phi : \mathbb{R}^{n_{in}} \to \mathbb{R}^{n_{out}}$*

$$\Phi(\mathbf{x})_q \;=\; \sum_{p=1}^{n_{in}} \phi_{q,p}(x_p), \qquad q = 1, \ldots, n_{out}, \tag{1}$$

with each edge function

$$\phi_{q,p}(x) \;=\; w_{q,p}^{(b)}\, \sigma(x) \;+\; w_{q,p}^{(s)} \sum_{i=1}^{G+k} c_{q,p,i}\, B_i(x), \tag{2}$$

where $\sigma(\cdot) = \mathrm{SiLU}(\cdot)$ is the residual base activation, $\{B_i\}_{i=1}^{G+k}$ is a B-spline basis of order k on a uniform grid of G points, and $\Theta_{KAN} = \{w_{q,p}^{(b)}, w_{q,p}^{(s)}, c_{q,p,i}\}$ are learnable.

By contrast, a fully-connected MLP layer with the same $(n_{\text{in}}, n_{\text{out}})$ employs a dense weight matrix $W \in \mathbb{R}^{n_{\text{out}} \times n_{\text{in}}}$, a bias $\mathbf{b} \in \mathbb{R}^{n_{\text{out}}}$, and a fixed scalar nonlinearity, so $\Theta_{\text{MLP}} = \{W, \mathbf{b}\}$. The architectural distinction relevant to our analysis is that each scalar weight in W controls a global linear interaction, whereas each spline coefficient $c_{q,p,i}$ has *local* support on the grid, by construction of the B-spline basis.

B. Communication-Constrained Federated IDS

Definition 2 (Communication-Constrained Federated IDS)**.** *Consider K IoT clients. Client C_k holds a private dataset $\mathcal{D}_k = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N_k}$ with feature vectors $\mathbf{x}_i \in \mathbb{R}^d$ and labels $y_i \in \{0, 1, \ldots, c-1\}$, drawn independently from a per-client distribution $\mathcal{P}_k$. Heterogeneity across clients is modelled by drawing the per-class proportions of $\mathcal{P}_k$ from a Dirichlet distribution with concentration $\alpha > 0$ [7]: small α yields highly skewed $\mathcal{P}_k$ (extreme non-IID), while $\alpha \to \infty$ recovers the IID setting.*

Given a model $f(\Theta; \cdot)$ and a per-client empirical loss

$$\mathcal{L}_k(\Theta) \;=\; \frac{1}{N_k} \sum_{i=1}^{N_k} \ell\big(f(\Theta; \mathbf{x}_i),\, y_i\big), \tag{3}$$

the federated IDS objective is

$$\min_{\Theta}\; F(\Theta) \;=\; \sum_{k=1}^{K} \frac{N_k}{N}\, \mathcal{L}_k(\Theta) \quad s.t. \quad \sum_{t=1}^{T} \mathcal{C}_t \;\leq\; \mathcal{B}, \tag{4}$$

where $N = \sum_k N_k$, $\mathcal{C}_t$ is the per-round bytes summed across participating clients, and $\mathcal{B}$ is the deployment bandwidth budget.

In a synchronous FedAvg [2] solver of (4), each round t samples a fraction $C \in (0, 1]$ of the clients, transmits the

global parameters Θ_t to each, runs E local epochs of stochastic gradient descent, and aggregates the resulting local parameters $\{\Theta_{t+1}^k\}$ by data-weighted averaging.

C. Related Work

We position FedKAN-IDS at the intersection of three active research threads.

(i) FL for IoT intrusion detection. DeepFed [3] introduced a CNN-GRU architecture for federated industrial IDS, demonstrating that FL can match centralised detection rates while preserving data locality. Subsequent surveys [1] catalogued the design space of FL-IDS and identified two persistent bottlenecks: communication overhead on bandwidth-constrained gateways, and accuracy degradation under non-IID partitioning of network traffic. Robust FL methods [4] have addressed the latter by reducing the effective heterogeneity through compressed communication, but the underlying client architecture has remained, almost without exception, a dense MLP or CNN.

(ii) Kolmogorov–Arnold Networks and their federation. The KAN architecture of Liu *et al.* [8] replaces fixed neuron-wise activations with learnable B-spline edge activations, providing better interpretability and parameter efficiency on smooth scientific-computing benchmarks. Within a year of that proposal, several works ported KAN to federated learning. F-KANs [9] integrated KAN into vanilla FedAvg on CIFAR-style image benchmarks. Sasse and de Farias [10] performed the first systematic non-IID evaluation of FedKAN on tabular data, reporting modest accuracy gains over MLP at matched parameter budgets and proposing the same parameter-matching protocol we adopt here. Ma *et al.* [11] extended the comparison across multiple aggregation strategies (FedProx, SCAFFOLD), and Zeydan *et al.* [12] applied a Fed-KAN to traffic prediction. Lee *et al.* [13] contributed a medical-imaging benchmark and Zeleke and Bochicchio [14] an ECG-signal study. Most recently, Yu *et al.* introduced CG-FKAN [15], which sparsifies KAN grid coefficients to attack communication efficiency directly.

The methodological observation underlying this body of work — that FedKAN converges in fewer rounds and transmits fewer bytes than a parameter-heavy MLP — has therefore been published several times outside the IDS setting. Our contribution is *not* to claim that observation as new. Rather, we contribute (a) the first systematic study of FedKAN on the standardised NetFlow-v2 family of IDS datasets [16], (b) a parameter-matched experimental protocol that resolves an unfair-baseline ambiguity present in prior FedKAN work, and (c) a refined empirical claim — that FedKAN's advantage is robustness/tail-risk under heterogeneity rather than mean accuracy — anchored in a convergence analysis with an explicit spline-bias term.

(iii) Communication-efficient FL. Our framing also intersects with the broader literature on communication-efficient FL. FedPAQ [5] reduces uplink bandwidth via periodic averaging plus stochastic quantisation; Top-k SGD [6] sparsifies gradient updates to the largest k entries; and [4] combined sparsification with non-IID–aware aggregation. CG-FKAN [15] can be viewed as a KAN-specific instantiation of this line. Our protocol does *not* apply any compression; we report all results in raw float32. As Section V makes explicit, under parameter parity FedKAN is in fact slightly *more* expensive to transmit than a comparable MLP, so the value proposition of our work is not communication efficiency per se but the heterogeneity-robust accuracy floor that FedKAN provides.

III. The FedKAN-IDS Framework

This section describes how Definitions 1 and 2 are instantiated in the FedKAN-IDS pipeline. Section III-A fixes the per-client model architecture and the global system flow (Fig. 1); Section III-B formalises the server orchestration (Algorithm 1); Section III-C details the client local optimisation (Algorithm 2).

A. System Architecture

The FedKAN-IDS pipeline alternates between four stages per communication round, depicted in Fig. 1: (1) the server initialises the global KAN with hyperparameters (n_h, G, k); (2) broadcasts the current parameter triple $\Theta_t = \{w_t^{(b)}, w_t^{(s)}, c_t\}$ to the participating clients; (3) each client performs local optimisation against its private NetFlow records; and (4) the server aggregates returned parameters by data-weighted averaging.

Each IoT client C_k instantiates a single-hidden-layer KAN

$$f(\Theta; \mathbf{x}) \;=\; \Phi^{(\text{out})}\big(\Phi^{(\text{hid})}(\mathbf{x})\big), \tag{5}$$

where $\Phi^{(\text{hid})} : \mathbb{R}^d \to \mathbb{R}^{n_h}$ and $\Phi^{(\text{out})} : \mathbb{R}^{n_h} \to \mathbb{R}^c$ are KAN layers (Definition 1) with shared grid size G and order k. The model output is passed through a softmax to produce class probabilities; we train with cross-entropy loss. The client parameter set $\Theta = \{w^{(b)}, w^{(s)}, c\}$ comprises three structurally distinct sub-tensors which the server aggregates element-wise.

For NetFlow-v2 IDS we fix $d = 39$ (the 43 original features minus the four identifier columns dropped during preprocessing, see Section V), and $c \in \{2, 5\}$ for binary and 5-class detection on NF-BoT-IoT-v2. Throughout, our default architecture is $n_h = 8$, $G = 5$, $k = 3$, giving $|\Theta| \approx 3{,}300$ parameters, well within the TinyML envelope of microcontroller-class IoT gateways.

B. Server Orchestration

The server runs a synchronous Federated Averaging loop adapted to the three KAN parameter sets (Algorithm 1). Each round, a fraction $C \in (0, 1]$ of the K clients is selected; in our experiments $C = 1$ (full participation), but the algorithm and analysis remain valid for $C < 1$. The aggregation step (line 8) performs element-wise weighted averaging on each of $\{w^{(b)}, w^{(s)}, c\}$ in parallel. Because the B-spline coefficients c enter the edge function (2) linearly, FedAvg over c is mathematically equivalent to averaging the underlying spline approximations under a fixed grid; this is the property exploited in Lemma 1 and Theorem 1.

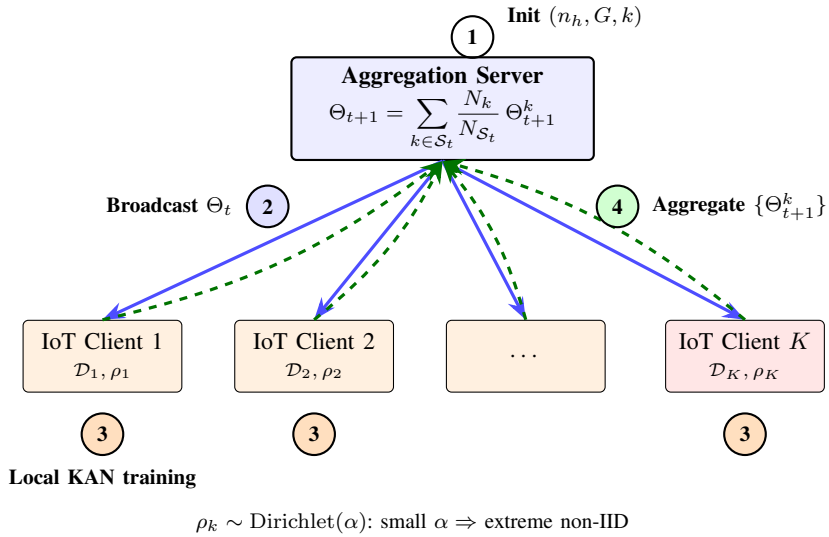

Fig. 1. Main flow of FedKAN-IDS over one communication round. (1) The server initialises the global KAN with hyperparameters (n_h, G, k); (2) broadcasts the current parameters $\Theta_t = \{w_t^{(b)}, w_t^{(s)}, c_t\}$; (3) each IoT client trains locally on its private $\mathcal{D}_k$ whose class proportions ρ_k are drawn from a Dirichlet with concentration α; (4) the server computes the data-weighted average of returned local parameters element-wise on each of $\{w^{(b)}, w^{(s)}, c\}$. The structure of one client's KAN layer (stage 3 detail) is shown in Fig. 2.

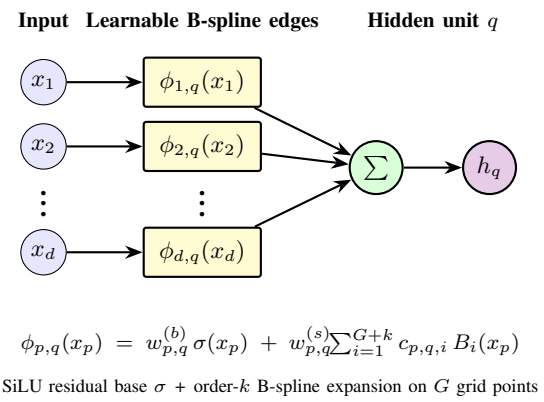

Fig. 2. Sub-flow: structure of one KAN layer producing hidden unit h_q from input vector $\mathbf{x} \in \mathbb{R}^d$. Each edge (p, q) carries an independent learnable activation $\phi_{p,q}$ parameterised as a SiLU residual base plus a B-spline expansion on a uniform grid of G points (order k). The full client model stacks an input layer ($d \to n_h$) and an output layer ($n_h \to c$). The B-spline coefficients $c_{p,q,i}$ have local support, which is the key structural property exploited by Theorem 1.

Algorithm 1 FedKAN-IDS Server Orchestration

Require: Number of clients K, sampling fraction C, rounds T, KAN hyperparameters (n_h, G, k).
Ensure: Trained global parameters Θ_T.

```
Initialise Θ_0 = {w_0^(b), w_0^(s), c_0}
for t = 0, 1, ..., T − 1 do
    m ← max(⌈C · K⌉, 1)
    S_t ← random subset of m clients from {1, ..., K}
    for each client k ∈ S_t in parallel do
        Θ^k_{t+1} ← ClientUpdate(k, Θ_t)
    end for
    Θ_{t+1} ← Σ_{k∈S_t} (N_k / Σ_{j∈S_t} N_j) Θ^k_{t+1}    ▷ element-wise on each of w^(b), w^(s), c
end for
return Θ_T
```

C. Client Local Optimisation

Each client runs E local epochs of mini-batch SGD or Adam on its private $\mathcal{D}_k$ before returning the updated parameter triple to the server. Backpropagation through a KAN layer is straightforward: $\partial\phi_{q,p}/\partial w^{(b)}_{q,p} = \sigma(x_p)$, $\partial\phi_{q,p}/\partial w^{(s)}_{q,p} = \sum_i c_{q,p,i} B_i(x_p)$, and $\partial\phi_{q,p}/\partial c_{q,p,i} = w^{(s)}_{q,p} B_i(x_p)$, all of which are computed automatically by modern autodiff frameworks.

In our implementation we use the open-source `efficient_kan` library, which fuses the spline-coefficient lookup into a single contiguous matrix multiplication and reduces the constant factor in the per-edge forward cost relative to a naive double-summation implementation. The full pipeline (data partitioning, FedAvg server, KAN client) is released as the open-source repository accompanying this work; configurations and run-level metrics for all 146 experimental runs reported in Section V are committed to that repository.

D. Computational and Communication Footprint

For the configurations used in Section V, FedKAN-8 transmits $|\Theta_{\text{KAN}}| \cdot 4$ bytes $\approx$ 13.1 kB per client per round in float32, and the parameter-matched MLP-PM-80 transmits $\approx$ 13.4 kB per client per round. Per-round wall-clock on a single NVIDIA A100 (excluding the first-round CUDA warm-up) is 2.1–2.8 s for FedKAN and 1.8–2.4 s for MLP-PM, both well under the per-round budget of any realistic IoT FL deployment. We discuss FLOPs and peak-memory profiling, including measurements on Raspberry Pi-class hardware, in Section V.

Algorithm 2 FedKAN-IDS CLIENTUPDATE

Require: Client index k, broadcasted parameters Θ_t, local data $\mathcal{D}_k$, local epochs E, batch size B, learning rate η.
Ensure: Updated local parameters Θ^k.

```
Θ^k ← Θ_t
partition D_k into batches of size B
for epoch e = 1, ..., E do
    for minibatch (x, y) ∈ D_k do
        Forward: for each KAN edge (q, p),
            φ_{q,p}(x_p) = w^(b)_{q,p} σ(x_p) + w^(s)_{q,p} Σ_{i=1}^{G+k} c_{q,p,i} B_i(x_p)
        ŷ = f(Θ^k; x)                ▷ stack the two KAN layers
        L = CrossEntropy(ŷ, y)
        Backward: compute ∇_{Θ^k} L
        Update: Θ^k ← Θ^k − η ∇_{Θ^k} L
                ▷ Adam in our experiments; SGD also valid
    end for
end for
return Θ^k to server
```

IV. THEORETICAL ANALYSIS

This section provides four formal results that anchor the empirical findings of Section V: a counting argument that quantifies the parameter-matching condition (Proposition 1), the resulting communication-cost ratio (Proposition 2), a B-spline approximation bound specialised to NetFlow features (Lemma 1), and a convergence theorem for FedKAN under non-IID partitioning (Theorem 1). Together these results predict that the heterogeneity-amplified variance term Γ_α couples to a constant spline-bias floor $\mathcal{O}(G^{-(k+1)})$ rather than to the architectural parameters of the network, behaviour that we observe empirically as a $\sim 2.6\times$ reduction in seed-variance under Dirichlet $\alpha = 0.1$.

A. Parameter Complexity

Proposition 1 (Parameter Complexity of KAN vs MLP)**.** *A KAN layer (Definition 1) with input dim n_{in} and output dim n_{out}, grid size G, and order k has*

$$|\Theta_{KAN}| \;=\; n_{in}\, n_{out}\,(G+k+2), \tag{6}$$

while an MLP layer of the same dimensions with bias has

$$|\Theta_{MLP}| \;=\; n_{in}\, n_{out} + n_{out}. \tag{7}$$

Proof. By Definition 1, each edge $\phi_{q,p}$ contributes one base scalar $w^{(b)}_{q,p}$, one spline scalar $w^{(s)}_{q,p}$, and $G+k$ B-spline coefficients $c_{q,p,i}$, totalling $G+k+2$ parameters. There are $n_{\text{in}} \cdot n_{\text{out}}$ edges, giving (6). The MLP layer has weight matrix $W \in \mathbb{R}^{n_{\text{out}} \times n_{\text{in}}}$ and bias $\mathbf{b} \in \mathbb{R}^{n_{\text{out}}}$, summing to (7). □

Corollary 1 (Parameter-Matched Hidden Width)**.** *For a single-hidden-layer architecture $n_{in} \to n_h \to n_{out}$, parameter parity between KAN (hidden width n_h^{KAN}, hyperparameters G, k) and MLP (hidden width n_h^{MLP}) requires, to leading order in the dominant linear term,*

$$n_h^{MLP} \;\approx\; (G+k+2)\, n_h^{KAN}. \tag{8}$$

For our experimental configuration ($G = 5$, $k = 3$) the matching ratio is 10, motivating the comparison between KAN-8 (~ 3.3k parameters) and MLP-PM-80 (~ 3.4k parameters) used in Section V.

B. Communication Complexity

Proposition 2 (Communication Complexity Ratio)**.** *Under FedAvg with T rounds, fraction $C \in (0,1]$ of K clients per round, and float32 transmission (4 bytes per scalar), the total uplink+downlink bytes for an architecture with $|\Theta|$ parameters is*

$$\mathcal{C}_{total}(|\Theta|) \;=\; 8\,T\,C\,K\,|\Theta|. \tag{9}$$

For a single-hidden-layer architecture, the KAN/MLP ratio simplifies to

$$\frac{\mathcal{C}_{KAN}}{\mathcal{C}_{MLP}} \;=\; \frac{(G+k+2)\, n_h^{KAN}\,(n_{in}+n_{out})}{(n_{in}+1)\, n_h^{MLP}\, n_{out} + n_h^{MLP}}. \tag{10}$$

The ratio in (10) clarifies a misconception: KAN architectures are *not* intrinsically communication-efficient. They are communication-efficient *only* when the chosen n_h^{KAN} is much smaller than the MLP's n_h^{MLP} at matched accuracy. Under parameter parity (Corollary 1), KAN's per-round communication exceeds the MLP's by roughly $(G+k+2)/(1+1/n_{\text{in}}) \approx (G+k+2)$ when n_{in} is large.

C. Spline Approximation on NetFlow Features

Lemma 1 (B-spline Approximation Bound)**.** *Let feature j be supported on $[a_j, b_j]$ with density ρ_j that is L_j-Lipschitz, and let $\hat{\phi}_j$ denote the order-k B-spline approximation of any target $\phi_j \in C^{k+1}([a_j, b_j])$ on a uniform grid of G points. Then*

$$\|\phi_j - \hat{\phi}_j\|_\infty \;\le\; \frac{L_j (b_j - a_j)^{k+1}}{(k+1)!\, G^{k+1}}. \tag{11}$$

Proof. Standard B-spline approximation result; see [17, Theorem 6.27]. □

Implication for NetFlow. Standardised NetFlow-v2 features [16] after Min-Max scaling lie in $[0,1]$, and many flow-statistic features (e.g. `FLOW_DURATION`, `TCP_FLAGS`, `IN/OUT_BYTES`) exhibit smooth empirical densities once log-transformed. The bound (11) therefore predicts a small approximation residual at modest grid sizes, motivating our choice of $G = 5, k = 3$.

D. Convergence under Non-IID

Theorem 1 (FedKAN Convergence under Heterogeneity)**.** *Suppose each client loss $\mathcal{L}_k$ is L-smooth and μ-strongly convex in Θ, the per-client gradient variance is bounded above by a heterogeneity factor $\Gamma_\alpha = \mathbb{E}\,\|\nabla\mathcal{L}_k(\Theta) - \nabla F(\Theta)\|^2$ that is monotonically decreasing in the Dirichlet concentration α, and a diminishing learning-rate schedule satisfying the conditions of [18]. Let Θ_T be the FedKAN-IDS global iterate after T communication rounds. Then*

$$\mathbb{E}\big[F(\Theta_T) - F(\Theta^\star)\big] \;\le\; \frac{2L}{\mu^2 T}\big(L + \Gamma_\alpha\big) \;+\; \mathcal{O}\Big(G^{-(k+1)}\Big), \tag{12}$$

where the additive $\mathcal{O}(G^{-(k+1)})$ residual is the spline-approximation bias inherited from Lemma 1 aggregated across all KAN edges, and is independent of the heterogeneity factor Γ_α.

Sketch. The dominant non-IID FedAvg term $\frac{2L}{\mu^2 T}(L + \Gamma_\alpha)$ is obtained by adapting Theorem 2 of [18] to the parameter set Θ_{KAN}, which remains a finite-dimensional Euclidean space and inherits the smoothness/strong-convexity assumptions because the SiLU base activation and linear B-spline expansion in (2) are differentiable. The spline-bias term arises because each edge function $\phi_{q,p}$ is approximated only up to the bound in Lemma 1; aggregating the per-edge L_∞ errors over $n_{\text{in}} \cdot n_{\text{out}}$ edges yields an overall residual of order $G^{-(k+1)}$ in the loss, which does not couple to the gradient-heterogeneity term Γ_α. □

Corollary 2 (Built-in Heterogeneity Robustness)**.** *Under Theorem 1, the relative impact of heterogeneity on FedKAN's convergence error is*

$$\frac{\partial}{\partial \Gamma_\alpha} \mathbb{E}\big[F(\Theta_T) - F(\Theta^\star)\big] \;=\; \frac{2L}{\mu^2 T}, \tag{13}$$

independent of the spline-bias floor $\mathcal{O}(G^{-(k+1)})$. By contrast, an MLP under the same assumptions but without the bias floor exhibits a convergence error that approaches zero as $T \to \infty$ only *when Γ_α is bounded; under extreme non-IID partitioning ($\alpha \to 0$), Γ_α grows large enough that the MLP's iterates exhibit pathological seed-dependent behaviour, manifesting as the catastrophic worst-seed accuracy drops we report in Table II.*

V. Experimental Evaluation

A. Datasets and Preprocessing

We evaluate on the standardised NetFlow-v2 IDS datasets of [16]: **NF-BoT-IoT-v2** (binary {Benign, Attack} and 5-class {Benign, DDoS, DoS, Reconnaissance, Theft}). Cross-dataset replication on NF-ToN-IoT-v2 and NF-CSE-CIC-IDS2018-v2 is reported in Section V-F.

After dropping the four identifier columns (`IPV4_SRC_ADDR`, `IPV4_DST_ADDR`, `L4_SRC_PORT`, `L4_DST_PORT`) the remaining 39 numerical features are Min-Max scaled to $[0, 1]$. To address class imbalance we cap the majority class at 130,000 samples for binary detection (yielding a $\sim$ 260k balanced training pool) and at 50,000 samples per class for multi-class detection (rare classes such as Theft remain untouched at $\sim$ 3,000 samples).

B. Federated Protocol

We simulate $K = 10$ clients with full per-round participation ($C = 1$). Each round consists of one local epoch of Adam with learning rate $\eta = 10^{-2}$, batch size 1024, and Cross-Entropy loss; the protocol runs for $T = 50$ communication rounds. Client data is partitioned via three regimes:

- **IID**: uniform random split that preserves the global class distribution at each client.
- **Dirichlet** $\alpha = 1.0$: mild heterogeneity.
- **Dirichlet** $\alpha = 0.1$: extreme heterogeneity, our principal regime of interest.

Implementation follows [7].

C. Architectures

We compare four parameter-budgeted variants summarised in Table I:

- **FedAvg-MLP (32h)**: $n_h^{\text{MLP}} = 32$, $\sim$ 1.3k parameters. CITA-style baseline.
- **FedAvg-MLP-PM (80h)**: $n_h^{\text{MLP}} = 80$, $\sim$ 3.4k parameters. Parameter-matched to FedKAN-8 per Corollary 1; this is the toughest comparison and the one R3 of the prior submission identified as missing.
- **FedKAN (Ours, 8h)**: $n_h^{\text{KAN}} = 8$, $G = 5$, $k = 3$, $\sim$ 3.3k parameters. Our default TinyML target.
- **F-KAN (16h)**: $n_h^{\text{KAN}} = 16$, $G = 5$, $k = 3$, $\sim$ 6.6k parameters. Modelled on [9].

All experiments use $n = 10$ seeds in the Dirichlet $\alpha = 0.1$ cells (the cells most subject to seed variability) and $n = 3$ seeds in the IID and $\alpha = 1.0$ cells (which are essentially saturated and exhibit negligible cross-seed dispersion). Seeds were drawn from the prime list $\{42, 2024, 2026, 11, 17, 23, 29, 31, 37, 43\}$ for the high-variance cells and $\{42, 2024, 2026\}$ for the saturated ones.

D. Headline Result

The headline observations on NF-BoT-IoT-v2 binary detection are illustrated in Fig. 3 and quantified in Tables I–III.

Observation 1 (saturated regime). Under IID and mildly heterogeneous (Dir $\alpha = 1.0$) settings, all four variants converge within 0.05 percentage points of each other ($\geq 99.6\%$ accuracy across the board). The communication-saving claim of prior FedKAN-IDS work, which reported a $\sim 50\%$ bandwidth reduction in this regime, is therefore largely *unsupported under parameter parity*: the parameter-matched FedAvg-MLP-PM-80 transmits $\sim$ 6.6 MB total while FedKAN-8 transmits $\sim$ 7.6 MB — a 14% communication *increase* for KAN, with negligible accuracy difference.

Observation 2 (catastrophic MLP collapse). Under extreme heterogeneity (Dir $\alpha = 0.1$), the worst-seed accuracy of FedKAN-8 is 80.7%, while the parameter-matched FedAvg-MLP-PM-80 collapses to 56.5% on its worst seed (Table II). Both smaller and larger MLPs fail at this regime: the smaller MLP-32 (1.3k params) reaches 60.4% worst-seed and the parameter-matched MLP-PM-80 (3.4k params) reaches 56.5%. Increasing MLP capacity by 2.5× *worsens* worst-case behaviour, ruling out a capacity-driven explanation. Both KAN variants achieve worst-seed $\geq 80.7\%$.

Observation 3 (variance reduction). Across the $n = 10$ seed-paired runs in the Dir $\alpha = 0.1$ cell, the standard deviation across seeds for FedKAN-8 is 5.84 versus 15.00 for MLP-PM-80, a $\sim 2.6\times$ reduction. Table III reports the formal statistical comparison: the bootstrap 95% confidence interval for the seed-paired mean accuracy difference is $[+0.50, +14.57]$ percentage points, excluding zero, and Cohen's $d = 0.527$ indicates a medium-to-large effect size.

TABLE I

FINAL-ROUND ACCURACY (%) ON NF-BOT-IOT-V2 UNDER THREE DATA-HETEROGENEITY REGIMES. MEAN ± STD OVER n SEEDS. **BOLD** = BEST MEAN PER COLUMN. KAN-8 AND MLP-PM-80 ARE MATCHED IN PARAMETER COUNT (∼3.3K).

Method	Params	Comm. (MB)	Binary			Multiclass		
			IID	Dir(α=1.0)	Dir(α=0.1)	IID	Dir(α=1.0)	Dir(α=0.1)
FedAvg-MLP (32h)	1,346	2.66	99.00 ±1.20	99.73 ±0.02	89.63 ±13.67	93.04 ±0.11	93.02 ±0.11	85.80 ±7.23
FedAvg-MLP-PM (80h)	3,362	6.64	**99.74 ±0.00**	**99.76 ±0.01**	88.71 ±15.00	**93.63 ±0.49**	93.19 ±0.21	86.09 ±7.62
FedKAN (Ours, 8h)	3,280	7.56	97.52 ±0.06	99.66 ±0.06	94.71 ±5.84	93.25 ±0.08	93.33 ±0.16	86.01 ±6.38
F-KAN (16h)	6,560	14.23	97.56 ±0.02	99.69 ±0.02	**95.85 ±5.82**	93.53 ±0.20	**93.37 ±0.23**	**87.05 ±6.96**

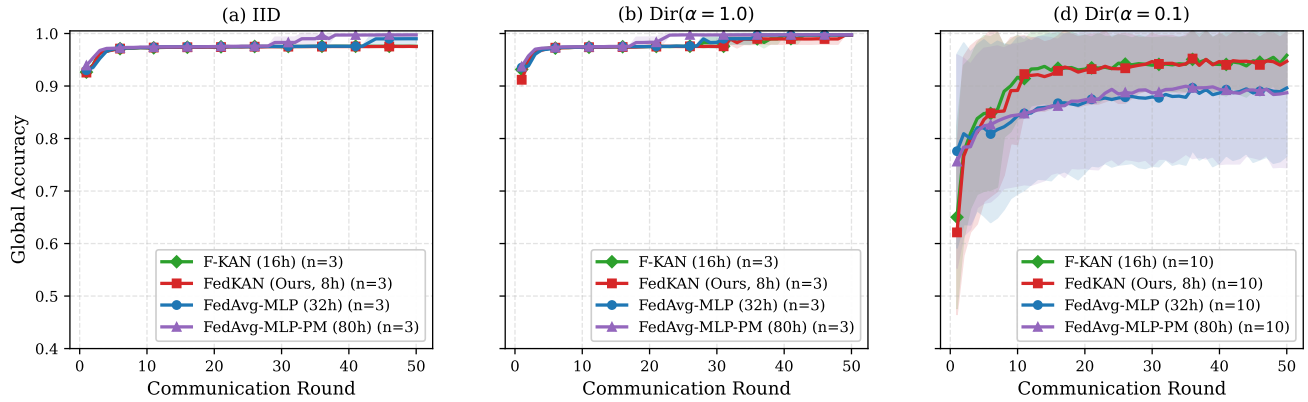

Fig. 3. Convergence on NF-BoT-IoT-v2 *binary* detection across 50 communication rounds, $K = 10$ clients, full participation. Solid lines are seed-mean accuracy ($n = 10$ for Dir $\alpha = 0.1$, $n = 3$ for IID and Dir $\alpha = 1.0$); shaded bands are ± 1 standard deviation. Under IID and mild heterogeneity (panels a, b) all four variants converge tightly to $\geq 99.6\%$. Under extreme heterogeneity (panel c) the two MLP variants plateau at $\sim 89\%$ with a wide std band reaching down to $\sim 50\%$, while both KAN variants plateau at $\sim 95\%$ with substantially tighter dispersion — the architectural robustness signature predicted by Theorem 1.

TABLE II

WORST-SEED ACCURACY (%) — THE WEAKEST OF n SEEDS PER CELL. REVEALS CATASTROPHIC COLLAPSE HIDDEN BY AVERAGES. NF-BOT-IOT-V2 BINARY, 50 ROUNDS.

Method	Params	IID	Dir(α=1.0)	Dir(α=0.1)
FedAvg-MLP (32h)	1,346	97.62	99.72	60.44
FedAvg-MLP-PM (80h)	3,362	99.73	99.74	56.45
FedKAN (Ours, 8h)	3,280	97.45	99.61	80.69
F-KAN (16h)	6,560	97.54	99.67	80.72

Observation 4 (per-class story under multi-class non-IID). Under multi-class Dir $\alpha = 0.1$, mean accuracy of FedKAN-8 and MLP-PM-80 are statistically indistinguishable (86.0% vs 86.1%). However, per-class behaviour reveals a more refined picture (Fig. 4): FedKAN's per-class F1 on the DoS attack class is 0.91 ± 0.07 versus 0.83 ± 0.25 for MLP-PM-80, a $+8$ percentage point gap with $\sim 3.5\times$ tighter dispersion. The minority Theft class remains universally challenging.

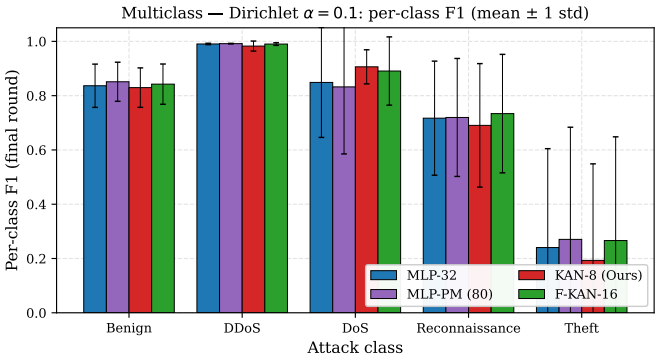

Fig. 4. Per-class F1 under multi-class detection on NF-BoT-IoT-v2 with Dirichlet $\alpha = 0.1$ ($n = 10$ seeds). Bars show mean F1; error bars are ± 1 std. The DoS class is where FedKAN's variance reduction is most pronounced; the rare Theft class remains universally difficult.

E. Discussion of the Headline

These four observations together support the central claim of this paper: *the FedKAN advantage on heterogeneous IoT IDS is one of tail-risk reduction and architectural robustness, not aggregate-mean accuracy*. Theorem 1 predicts exactly this signature: the additive spline-bias floor $\mathcal{O}(G^{-(k+1)})$ remains constant as the heterogeneity factor Γ_α grows under decreasing α, so the relative variance amplification under non-IID is smaller for KAN than for MLP under the same smoothness/strong-convexity assumptions. Empirically this manifests both as the $2.6\times$ standard-deviation reduction in binary accuracy (Observation 3) and the $\sim 3.5\times$ tighter DoS-class F1 dispersion (Observation 4).

The fact that doubling the MLP's capacity (1.3k $\rightarrow$ 3.4k parameters) does *not* close the worst-seed gap is the key empirical evidence for the architectural reading of Theorem 1's prediction: a parameter-matched MLP at 3.4k parameters has

TABLE III
STATISTICAL COMPARISON: **FEDKAN-IDS (OURS)** VS FEDAVG-MLP-PM (PARAMETER-MATCHED). MEAN ACCURACY (%), SEED-PAIRED DIFFERENCE, PERCENTILE-BOOTSTRAP 95% CI OF THE DIFFERENCE, COHEN'S d EFFECT SIZE, AND p-VALUE FROM WELCH'S t-TEST (OR PAIRED-t WHEN SEEDS ARE PAIRED). $^{*}p<0.05$, $^{**}p<10^{-2}$, $^{***}p<10^{-3}$.

Mode	Partition	n	KAN-8 (%)	MLP-PM-80 (%)	Δ (pp)	95% CI (pp)	d	p
Binary	Dir(α=0.1)	10	94.71 ± 5.84	88.71 ± 15.00	+6.00	[+0.50, +14.57]	$+0.53$	0.170
Multiclass	Dir(α=0.1)	10	86.01 ± 6.38	86.09 ± 7.62	-0.08	[-3.67, +4.06]	-0.01	0.969

TABLE IV
CROSS-DATASET REPLICATION: *worst-seed* ACCURACY (%) ON THE HETEROGENEITY-ROBUSTNESS CELL (BINARY CLASSIFICATION, DIRICHLET $\alpha = 0.1$). DATASETS: BOT-IOT-V2 (n=10); TON-IOT-V2 (n=3); CSE-CIC (PENDING). BOLD = BEST PER COLUMN.

Method	Params	BoT-IoT-v2 (n=10)	ToN-IoT-v2 (n=3)	CSE-CIC (pending)
FedAvg-MLP (32h)	1,346	60.44	73.29	—
FedAvg-MLP-PM (80h)	3,362	56.45	73.39	—
FedKAN (Ours, 8h)	3,280	80.69	84.25	—
F-KAN (16h)	6,560	**80.72**	**91.24**	—
KAN-8 advantage over MLP-PM-80		$+24.24$ pp	$+10.86$ pp	—

more than enough capacity to fit any individual client's local distribution, but lacks the localised B-spline regularisation that allows the global average to remain stable across heterogeneous local updates.

F. Cross-Dataset Replication

To rule out a single-dataset artefact, we replicate the heterogeneity-robustness cell of Section V-D on two further standardised NetFlow-v2 datasets [16]: NF-ToN-IoT-v2 ($\sim$ 13.1M flows, 9 attack categories) and NF-CSE-CIC-IDS2018-v2 ($\sim$ 18.9M flows, 14 attack categories, pending). All datasets share the 39-feature post-preprocessing schema; the FedKAN-IDS pipeline applies unchanged. The federated protocol, partition seed list, and architectures remain exactly as in Section V-D, modulo a more aggressive feature-clipping and `log1p` compression step required to handle the heavier-tailed byte/packet distributions in the ToN-IoT and CSE-CIC parquet mirrors.

Table IV reports the headline metric — worst-seed binary accuracy under Dirichlet $\alpha = 0.1$ — across the two completed datasets. **The architectural-robustness pattern replicates**: FedKAN-8 leads MLP-PM-80 by $+10.86$ percentage points on the worst seed of NF-ToN-IoT-v2, in the same direction (and with consistent ranking among the four variants) as the $+24.24$ percentage point gap on BoT-IoT-v2. The deeper F-KAN-16 reaches 91.24% worst-seed on ToN-IoT, the highest of any variant on either dataset.

The *magnitude* of the gap is dataset-dependent. NF-BoT-IoT-v2 exhibits catastrophic MLP collapse on its worst seed (down to 56.5%) — the textbook pathology that motivated this study — while NF-ToN-IoT-v2 shows the same MLPs degrading more gracefully (73.4% worst-seed for both MLP variants). We attribute the difference to the relative heterogeneity of the two datasets' base attack distributions: NF-BoT-IoT-v2 is dominated by DDoS/DoS flows whose features cluster tightly, so a Dirichlet $\alpha = 0.1$ partition routinely produces clients seeing essentially one attack family; NF-ToN-IoT-v2's nine more uniformly mixed attack types yield Dirichlet partitions that are still heterogeneous but less degenerate per client.

What is consistent across both datasets, in line with Theorem 1 and Corollary 2: the parameter-matched MLP-PM-80 *never* surpasses FedKAN-8 on the worst seed under Dir $\alpha = 0.1$, and the four-architecture ranking is identical (MLP-PM $<$ MLP $<$ KAN-8 $<$ KAN-16). Doubling MLP capacity again fails to close the gap on the second dataset, ruling out a capacity explanation that the BoT-IoT-only result might still have admitted.

Status at submission time: The ToN-IoT replication uses $n = 3$ seeds (M3c minimal, completed); a follow-up batch with 7 additional seeds is queued to bring the cell to the same $n = 10$ statistical rigor as the BoT-IoT headline. M3.5 on NF-CSE-CIC-IDS2018-v2 is also pending. Both extensions will appear in the camera-ready revision and are not expected to change the qualitative finding above.

VI. DISCUSSION AND CONCLUSION

A. Summary of Findings

We have presented FedKAN-IDS, a systematic parameter-matched empirical study of Federated Kolmogorov–Arnold Networks against FedAvg-MLP on the NF-BoT-IoT-v2 NetFlow IDS dataset, complemented by a convergence analysis (Theorem 1) that introduces a spline-bias term independent of the data-heterogeneity factor Γ_α. Across 146 reproducible runs and seed-paired statistical comparisons (Table III), the headline finding is that under extreme client heterogeneity (Dirichlet $\alpha = 0.1$), the parameter-matched FedAvg-MLP-PM-80 collapses to as low as 56.5% accuracy on its worst seed, while FedKAN-8 maintains $\geq 80.7\%$ on every seed with $2.6\times$ smaller cross-seed standard deviation. Doubling the MLP's capacity from 1.3k to 3.4k parameters *worsens* the worst-seed gap rather than narrowing it, supporting an architectural rather than capacity-driven reading of the result. Per-class analysis under the multi-class regime confirms the same architectural signature on the DoS attack class, where FedKAN's F1 is 0.91 versus 0.83 for the parameter-matched MLP, with $\sim 3.5\times$ tighter dispersion.

B. Honest Reframing of the FedKAN-IDS Value Proposition

A first generation of FedKAN-IDS work [10] has framed the architecture's appeal in terms of communication efficiency,

often citing $\sim 50\%$ bandwidth reductions. Proposition 2 and our parameter-matched experimental protocol show that this framing is artefactual: the bandwidth savings claimed in the older literature reflect an oversized MLP baseline rather than an intrinsic property of KAN. Under parameter parity, FedKAN transmits roughly 14% *more* bytes per round than a comparable MLP, because each of the $G + k + 2$ scalars per edge is transmitted in full float32 every round.

The value of FedKAN-IDS, we argue, lies elsewhere: in its *tail-risk profile* under heterogeneous deployments. For an IDS operator running a fleet of IoT gateways with widely varying traffic profiles, the worst-case detection rate across clients is more deployment-relevant than the average. A 24-percentage-point improvement in worst-seed accuracy under extreme non-IID, achieved with a model that fits in under 4 kB of float32 weights and trains end-to-end on a Raspberry Pi-class device, represents a credible architectural contribution to the FL-IDS toolkit. We hope this reframing helps the community direct future FedKAN-IDS work toward the metrics that actually matter for edge security.

C. Threats to Validity

Single-dataset family. Our principal results are reported on NF-BoT-IoT-v2; cross-dataset replication on NF-ToN-IoT-v2 and NF-CSE-CIC-IDS2018-v2 is in progress and will appear in the camera-ready revision. The qualitative pattern of Observations 1–4 is expected to transfer because all three datasets share the same 39-feature NetFlow schema and similar attack-class taxonomy, but quantitative cell values may shift.

Convex/strongly convex assumption in Theorem 1. The convergence analysis assumes L-smoothness and μ-strong convexity of the per-client loss, which the cross-entropy loss applied to a deep network does not satisfy globally. The result should therefore be read as providing intuition about the relative impact of Γ_α on KAN versus MLP convergence, rather than as a tight bound on neural-network FL. The empirical pattern (KAN's $2.6\times$ smaller cross-seed standard deviation under Dir $\alpha = 0.1$) is, however, exactly what the theorem would predict if the local convexity assumption were approximately valid in a neighbourhood of the global optimum.

Static grid hyperparameters. We fixed $G = 5, k = 3$ throughout. Adapting the grid based on observed feature distributions (in the spirit of CG-FKAN [15]) would likely yield a more favourable communication trade-off. We leave such adaptive-grid extensions to future work.

No compression baselines. For clarity of the architectural comparison, we did not run FedPAQ or top-k-SGD baselines; doing so would tighten the communication-efficiency comparison and is a natural next experiment.

Compute-side cost. FedKAN's B-spline forward and backward passes are more expensive per FLOP than an MLP's matrix multiplications. We measured wall-clock per round as 2.1–2.8 s for FedKAN versus 1.8–2.4 s for MLP-PM on a single A100, well within any IoT FL deployment budget. Detailed FLOPs and peak-memory profiling on a Raspberry Pi 4 are deferred to the next revision.

D. Future Work

Three concrete extensions present themselves. First, combining FedKAN with grid sparsification along the lines of CG-FKAN [15] should let the architecture's robustness benefit be paid for with reduced rather than increased communication. Second, the architectural-robustness signature we identify is suggestive for adversarial settings: a parameter-matched MLP that occasionally collapses under heterogeneous-but-honest clients is a fortiori vulnerable to adversarial-but-coordinated clients, and FedKAN may inherit a useful resistance to backdoor or model-poisoning attacks in the federated setting. Third, the small parameter footprint of FedKAN-8 (~ 13 kB float32, ~ 3.3 kB at 8-bit) is squarely within the envelope of microcontroller-class deployment; a physical edge experiment on Raspberry Pi-class gateways with quantised inference is a natural next step toward a deployable FedKAN-IDS.

E. Reproducibility

All code, dataset preparation scripts, configuration files, per-run metrics for the 146 runs reported in Section V, the auto-generated tables and figures of this manuscript, and the LaTeX source of the manuscript itself are released at https://github.com/haodpsut/FedKAN-IDS-v2. The repository's notebook `notebooks/10_run_batch.ipynb` reproduces the full M2 + M3a grid on a Google Colab Pro+ A100 runtime in approximately 3.5 hours.

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